

**STATEMENT OF WORK
IRRIGATION PUMP REPLACEMENT
JOINT BASE SAN ANTONIO, RANDOLPH AIR FORCE BASE
13 JANUARY 2025**

I. DESCRIPTION OF SERVICES

1. SCOPE OF WORK: Contractor shall replace the current irrigation pump #2 located at Building 1302 (Golf course pump station). Pump #2 is 20 years old and is no longer performing at peak efficiency. There are 2 pumps in the pump station (BLDG 1302) that are supposed to each pump 1160 gallons per minute (GPM) for a total system capability of 2320 GPM. Pump #1 was replaced in 2024 and is 84.3% efficient at 1160 gpm, however pump #2 is 60% efficient at 660 gpm. The Randolph Golf course requires this pump to be replaced to restore max efficiency to the irrigation pump station. Additionally, Randolph Golf Course needs to have a fully operational system during its peak season (May to October). The task must be completed no later than April 30, 2025. The start to finish time must be no more than 1 workday. The contractor will supply all materials related to this job to include, new pump, crane, pump shaft, check valves, hardware, etc. Below are the known steps for completing this task:

1. Isolate the pump station from electricity by shutting off the power to BLDG 1302.
2. Disconnect the electric lines to that supplies 480 volts to the 100 HP electric motor that drives the irrigation pump.
3. Uncouple the 100 HP motor from the pump shaft.
4. Disconnect the 8" check valve on the discharge side of the pump.
5. Use a crane to hoist the 100 HP motor off the pump shaft.
6. Use a crane to hoist the pump and pump shaft out of the pump station wet well through the roof of the pump station. The roof has a removable panel that makes this possible.
7. Use a crane to hoist the new pump and pump shaft through the roof and into the pump station wet well.
8. Fasten the pump shaft and 8" check valve to the discharge side of the pump.
9. Use a crane to hoist the 100 HP motor onto the pump shaft.
10. Couple the pump shaft to the 100 HP motor.
11. Connect the electric lines that supply the 100 HP motor with 480 volts.
12. Power the pump station back on.
13. Test the new pump to ensure it is fully operational.

2. PERIOD OF PERFORMANCE: Because Randolph Oaks Golf Course will be incapable of irrigation during the period of performance the contractor must complete the pump replacement within 1 workday so that the golf course is not without irrigation for longer than 24 hours.

3. PHYSICAL LOCATION: Building 1302 (Irrigation Pump Station)

4. SUPPLIES AND EQUIPMENT: The new pump must be identical to the current #1 pump that was installed this past April of 2024 so that both pumps have identical efficiencies. This is because of the way the pump station keeps up with the irrigation systems demand for irrigation water downstream. Below are the equipment and supplies required as well as the pump specifications:

1. 100HP Vertical turbine Pump, (84.3 Efficiency), rates for 1160 gpm @ 285' TDH
2. Cast Iron Bowls w/316 investment cast SS Impellers
3. 416SS Stainless Steel Shafting
4. 416SS Impeller Collets
5. Water lubricated line shaft bearings with maximum 5 ft. spacing
6. MCI custom leakless packing system designed to prevent water from leaking onto the pump skid
7. 8" MCI Fusion Bonded Epoxy coated internal/external fabricated steel discharge head
8. Threaded carbon steel column pipe, standard wall thickness, 5' ft sections
9. Stainless steel pump strainer
10. 8" grooved pump isolation valves with lever actuator
11. 8" grooved pump discharge flange
12. 8" pump silent check valve coated with fusion bonded epoxy and SS disc.
13. New hardware such as nuts, bolts, etc.

5. PAYMENT RATE: [REDACTED]

6. ACCEPTANCE CRITERIA: At the completion of the job the new pump shall provide a flow of 1160 gpm with no pump station faults and shall not leak water from any connections such as check valves, flanges, etc.

7. TOTAL NOT TO EXCEED: [REDACTED]

8. SPECIAL REQUIREMENTS: Our pump station is an MCI Flowtronex irrigation pump station so the contractor must be certified by MCI Flowtronex to replace / install any equipment within the pump station.

9. SAFETY The contractor shall ensure compliance with OSHA Safety and Health Regulation.

10. WASTE DISPOSAL AND ENVIRONMENTAL PROTECTION: At the end the work day, the contractor shall remove all residual equipment, components, materials, and/or debris generated and disposed of in the appropriate government containers.

5. SECURITY AND CLEAN-UP: All areas shall be cleaned of debris at the end of the working day and prior to final acceptance of the work.

6. WARRANTIES: The new pump must have a 1-year warranty of all parts and components. In addition there must be at least a 90 day warranty on workmanship to cover any issue that was deemed the fault of the installer such as unsecured electrical connections, etc.

7. COMPLETION OF WORK: The work will be considered complete when the current pump has been removed and replaced with the new pump, when the new pump has been tested and determined by the user and the installer to be operating properly and successfully.